Exp No: 25

PRODUCER–CONSUMER PROBLEM USING SYSTEMC

AIM:

To implement and simulate the Producer–Consumer problem using SystemC.

SOFTWARE REQUIRED:

Ubuntu / WSL Linux

g++ Compiler

CMake

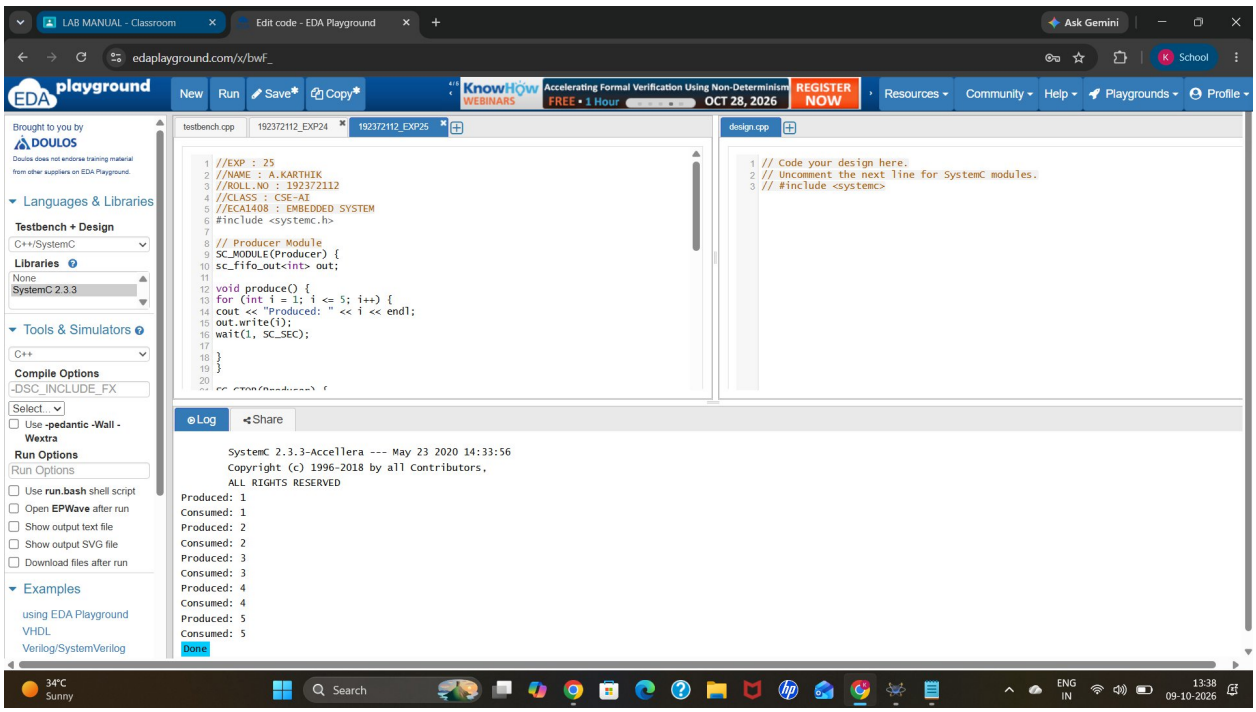
SystemC Library

PROGRAM:

```
#include <systemc.h>
// Producer Module
SC_MODULE(Producer) {
sc_fifo_out<int> out;
void produce() {
for (int i = 1; i <= 5; i++) {
cout << "Produced: " << i << endl;
out.write(i);
wait(1, SC_SEC);
}
}
SC_CTOR(Producer) {
SC_THREAD(produce);
}
};
// Consumer Module
SC_MODULE(Consumer) {
sc_fifo_in<int> in;
```

```
void consume() {
int value;
while (true) {
in.read(value);
cout << "Consumed: " << value << endl;
wait(1, SC_SEC);
}
}
SC_CTOR(Consumer) {
SC_THREAD(consume);
}
};
int sc_main(int argc, char* argv[]) {
sc_fifo<int> fifo(5);
Producer producer("Producer");
Consumer consumer("Consumer");
producer.out(fifo);
consumer.in(fifo);
sc_start(10, SC_SEC);
return 0;
}
```

OUTPUT:



RESULT:

Thus, the Producer–Consumer problem was successfully implemented and simulated using SystemC. Proper synchronization between concurrent processes was achieved using FIFO communication.